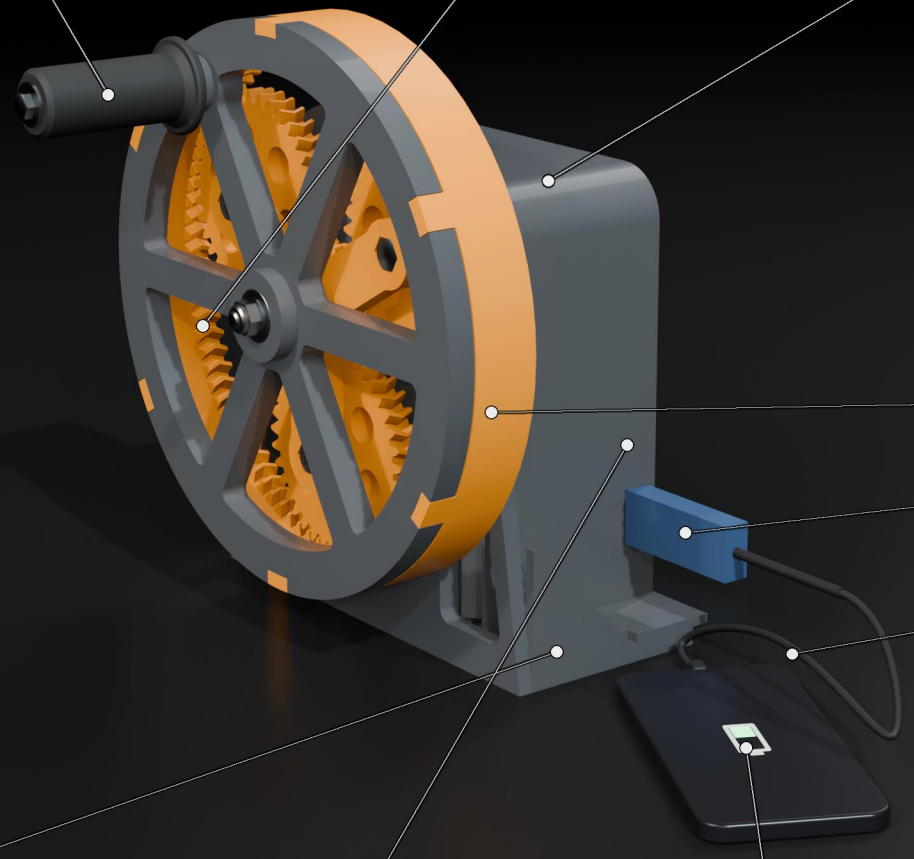
1 The machine

hand-crank phone charger, as built



grip
printed; turns free on an M6 x 90 post

hand wheel
printed, 236 mm; the ring's six dogs sit in its notches

wire cover
printed; clicks on behind the wall

ring gear
printed, 91 teeth, behind the wheel; its dogs drive it

USB meter
volts and amps on the way to the phone

phone lead

frame
printed; wall, base and five gussets in one piece

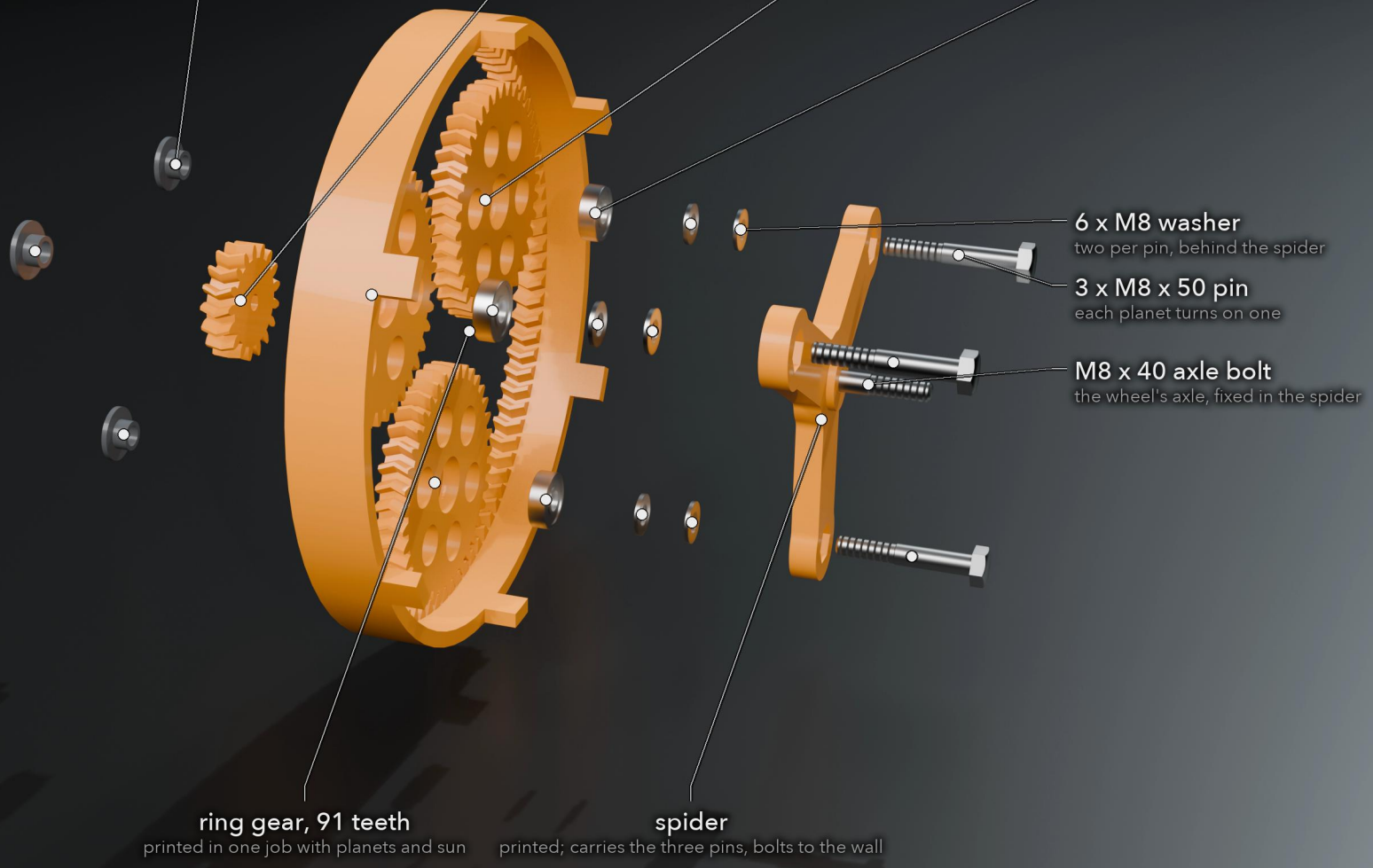
inside the cover
NEMA23 motor, 8 Schottkys, 1000 uF, 5 V converter

phone
charging at 5 V, up to 2 A

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3 The gear cartridge

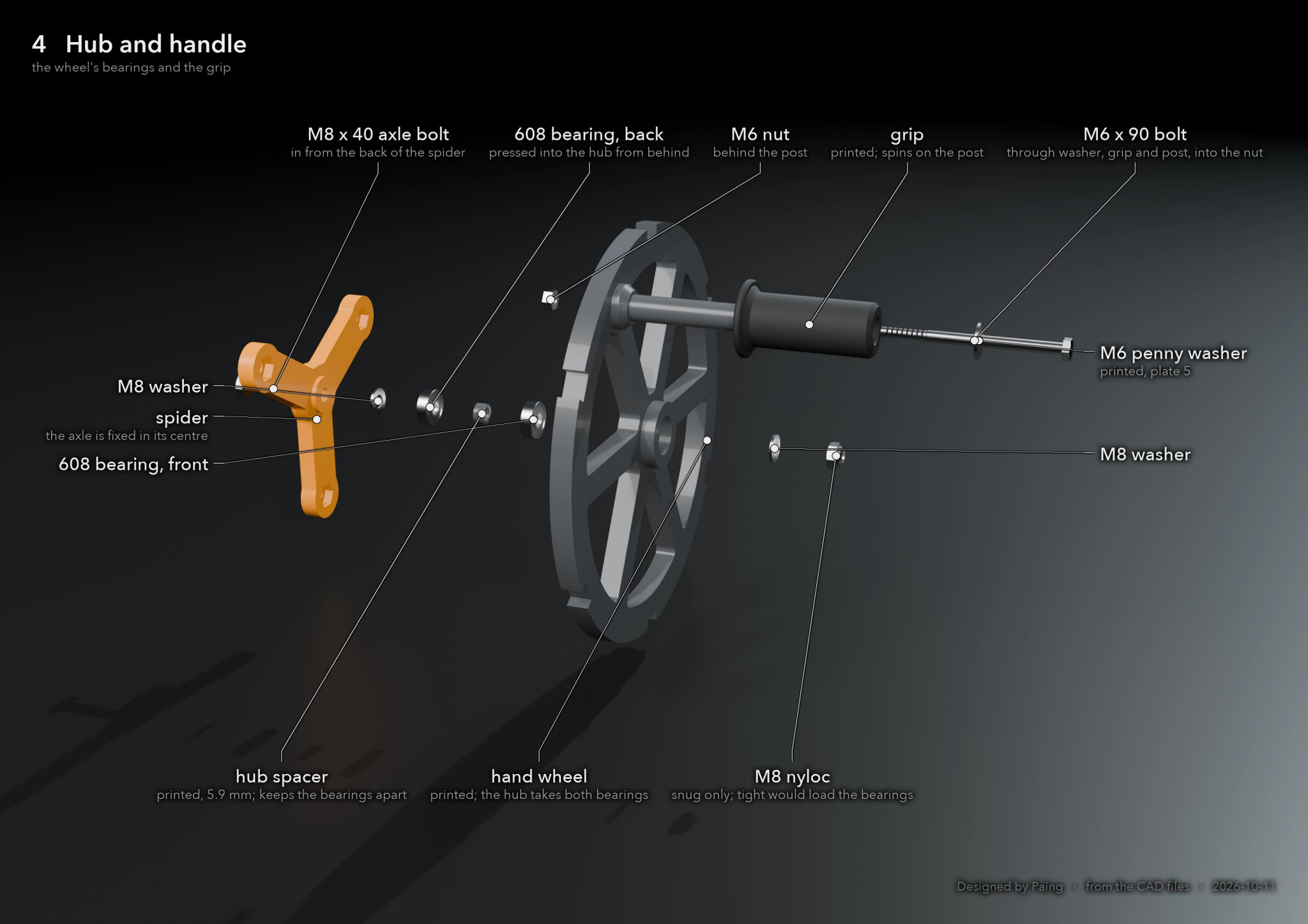
one planetary stage, 91 : 17 = 5.35 to 1



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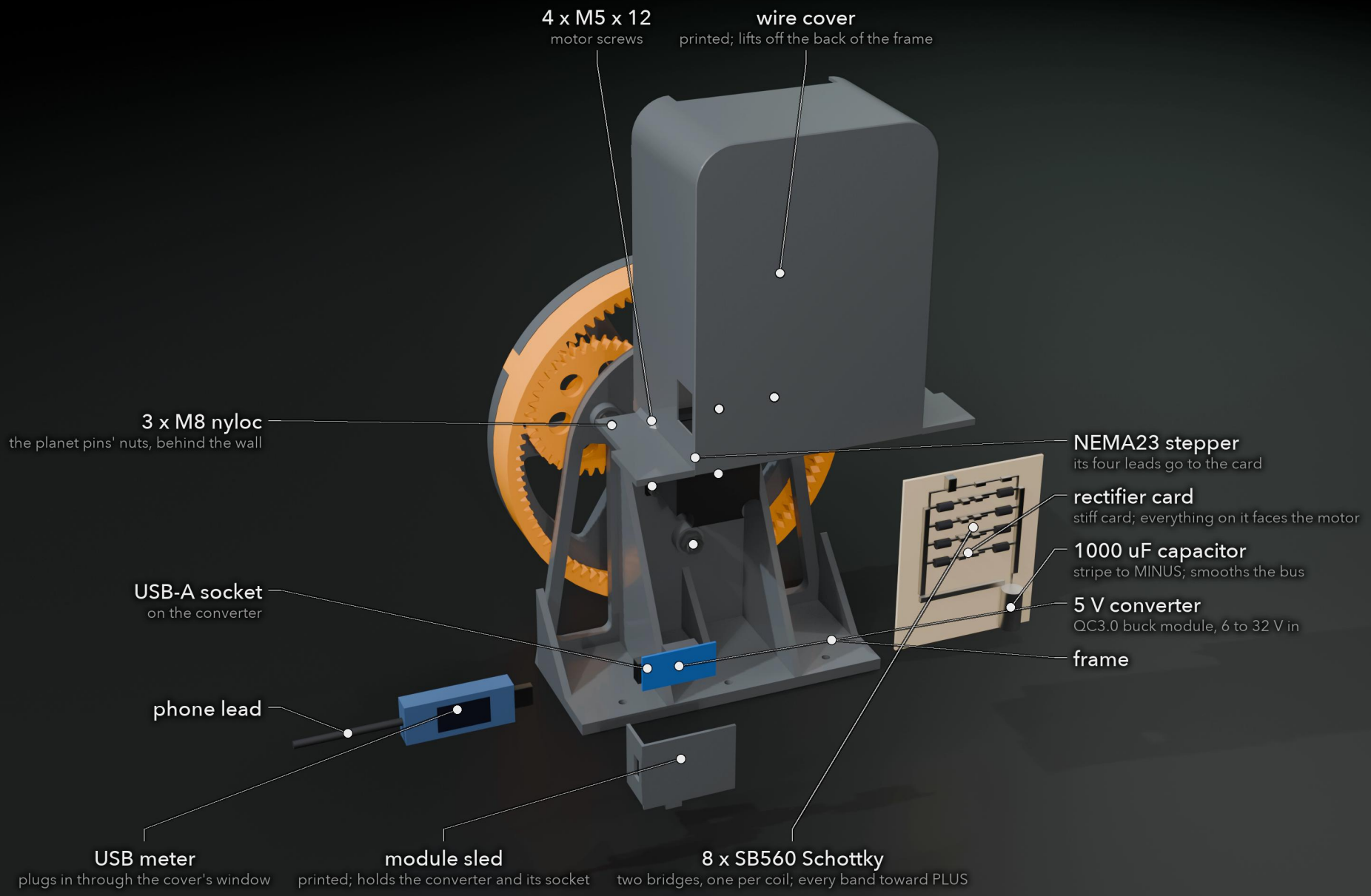
4 Hub and handle

the wheel's bearings and the grip



5 The electronics

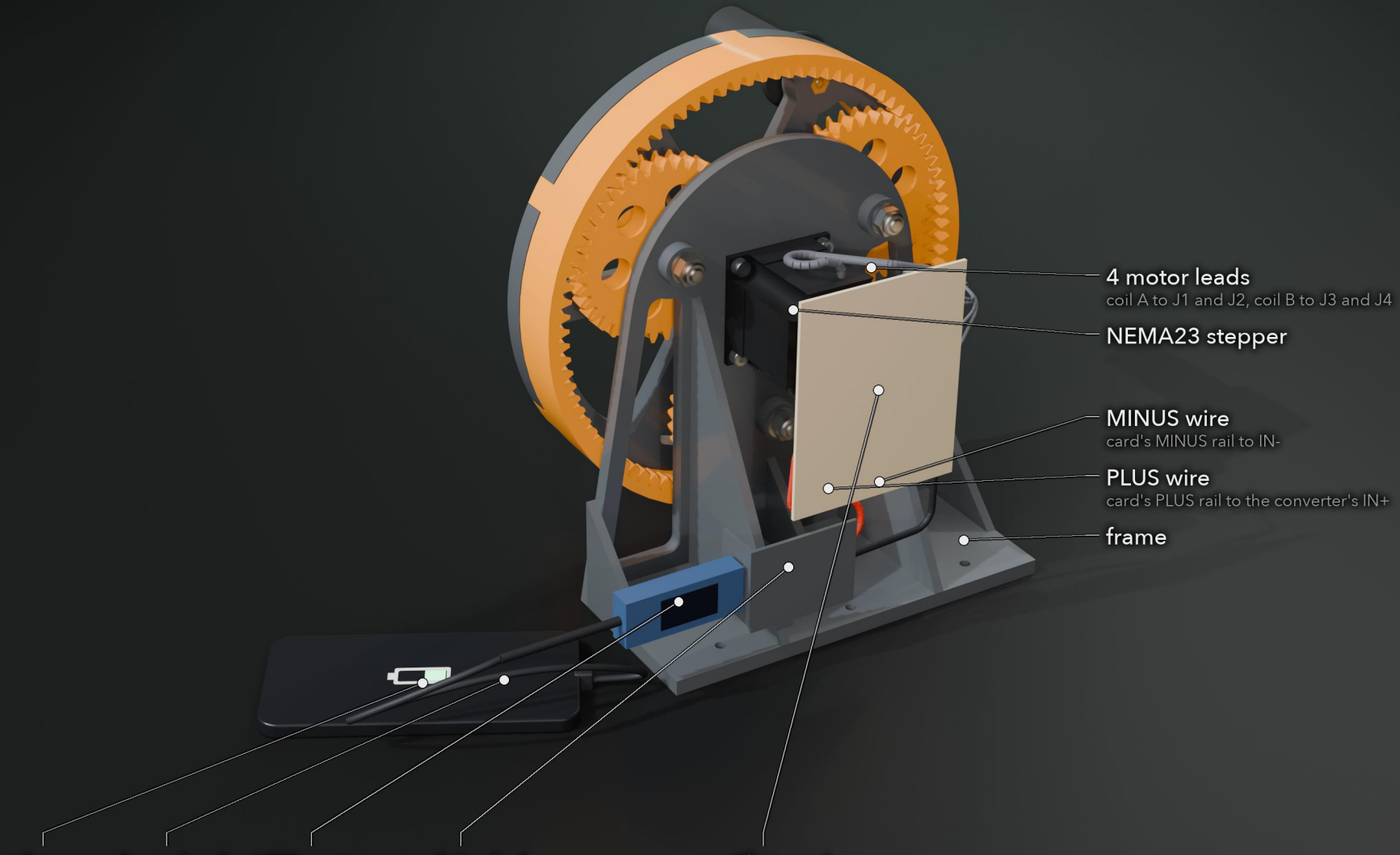
lifted out of the cover



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6 Electronics in place

cover off, everything wired



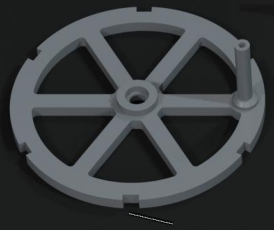
Designed by Paing · from the CAD files · 2026-10-11

7 The print files

the machine's own parts, as exported, in print orientation



gear set, print in place
plate 2: ring 91, three planets 37, sun 17, teeth on the bed



hand wheel
plate 3: back on the bed, post up



frame
plate 4: front face on the bed



wire cover shell
cover file 2: open side up



spider
plate 1



12 M8 + 1 M6 penny washer
plate 5, 1.6 mm



grip
plate 1: stands on its disc



3 x pin spacer
plate 1



hub spacer
plate 1